

Environmental Risk Assessment and Operational Decision Support System (Project Proposal)

Problem Context – summary:

Nowadays, the importance of sustainability and focusing on environmental factors has increased, especially with the 2030 vision. Organizations, and specifically contractors that work under dynamic environmental conditions, face many challenges. Such as, Decision makers struggle to interpret multiple indicators together, and their decisions are mostly based on personal judgment and not understanding the environmental factors; there is no unified risk level or documented decision logic.

Problem Statement:

The main problem is the big challenges that organizations and contractors face in sectors like real estate when the weather and environmental conditions are significant. Right now, the data we have is scattered, which makes it really hard to get clear insights into making the right decisions. This is especially clear when a real estate company starts a specialized project for the first time, like a coastal project where they don't have enough experience with the specific laws, regulations, and governance of that environment. Because there isn't a unified system that connects weather data with legal and operational rules, the decisions end up being based on personal judgment. This will lead to confusing procedures on site, also real safety risks, and the chance of breaking environmental laws or facing legal problems just because there was no proper documentation to justify why a decision was made, and if it's an accurate decision.

Target Users:

Our system is not for public users, but for internal organizational use. The system is designed for operational decision-makers, including Contractors, Operations Managers, Project Managers, Risk Management Officers, and many other decision-makers. Our prototype's main focus will be on contractors. **Why Contractors?** Because they are highly exposed to environmental risks, yet many decisions are still experience-based.

Business impact:

Starting a project without addressing the environmental factors will eventually lead to Safety risks, Operational disruptions, and inconsistent decisions that may affect the organization's workflow and cause financial loss. Also, not working according to the environment and sustainability regulations might affect the company's reputation.

Problem–Oriented Objectives:

1. To analyze environmental and weather hazards using a structured, comprehensive system.
2. Providing decision-makers with clear and understandable risk-based recommendations.
3. To allow decision-makers to make calculated risk decisions without needing expert knowledge in the environmental or weather fields.

The Project KPIs:

1. Effectiveness of the Overall Risk Score: The overall risk score's ability to accurately represent overall environmental risk and provide an accurate number by at least 90% for the classification.
2. Reliability of Risk Classification: Environmental conditions are at least 85% accurately classified into the correct risk levels (Low, Medium, High, Critical) using specified criteria.
3. Utilization of Non-Expert Users: 95% of the decision-makers without environmental experience can understand the system's results and implement recommendations to make accurate decisions.
4. System Response Time: The time required for the system to assess environmental conditions, create a risk score, and make operational recommendations in less than 2 minutes.

Scope of the Problem:

The scope covers risk assessment, decision support, and decision evaluation. The project excludes advanced environmental modeling, forecasts, machine learning approaches, real time system integration, and effective solutions.

Constraints and Assumptions:

Constraints:

- Data availability: Our system will use either simulated or public environmental data rather than real-time datasets.
- Technical scope: Advanced predictive analytics, machine learning models, and complex environmental simulations are outside the project scope.
- Limitations for implementation: The system will be developed as a prototype and not implemented in a real operational environment.

Assumptions:

- The environmental factors utilized by the system are effective for illustrating common operational risk situations.
- Decision-makers using the system may lack specialized environmental knowledge.
- The system will serve as a decision-support tool, not an alternative for expert judgment or regulatory authority.

“NOTE”

The source code and full technical documentation are currently not public because the project team is considering intellectual property registration.